Missing value (Before cleaning)

[5]
0s

```python
# Count missing values in each column
print(df.isna().sum())
```

```
restaurant       0
item             0
calories         0
cal_fat          0
total_fat        0
sat_fat          0
trans_fat        0
cholesterol      0
sodium           0
total_carb       0
fiber           12
sugar            0
protein          1
vit_a          214
vit_c          210
calcium        210
salad            0
dtype: int64
```

Check Duplicate Rows **(Before cleaning)**

[6]
✓ 0s

```
# checking duplicate rows
df.duplicated()[df.duplicated() == True]
```

	0
492	True
511	True

dtype: bool

- There is 2 duplicate rows, located at indices 492 and 511.

No. of Rows and Columns **(Before cleaning)**

[7]
✓ 0s

```python
# finding number of rows and columns
print("Shape (rows, columns): ", df.shape, "\n")

print("number of rows: ", df.shape[0])
print("number of columns: ", df.shape[1])
```

```
Shape (rows, columns):  (515, 17)

number of rows:  515
number of columns:  17
```

Remove Duplicate Rows **(After Cleaning)**

[11]
✓ 0s

```
# Remove duplicate rows
df = df.drop_duplicates()

# Check again
df.duplicated().sum()
```

```
np.int64(0)
```

✏️ Explanation:

- Duplicate rows were removed to avoid repeated data.
- This ensures more accurate analysis.

Handle Missing Values **(After cleaning)**

[12]
✓ 0s

```
# Fill missing values with mean for numeric columns
df.fillna(df.mean(numeric_only=True), inplace=True)

# Check again
df.isna().sum()
```

	0
restaurant	0
item	0
calories	0
cal_fat	0
total_fat	0
sat_fat	0
trans_fat	0
cholesterol	0
sodium	0
total_carb	0
fiber	0
sugar	0
protein	0
vit_a	0

[12]
0s

```
df.isna().sum()
```

	0
restaurant	0
item	0
calories	0
cal_fat	0
total_fat	0
sat_fat	0
trans_fat	0
cholesterol	0
sodium	0
total_carb	0
fiber	0
sugar	0
protein	0
vit_a	0
vit_c	0
calcium	0
salad	0

dtype: int64

Verify Cleaning **(After cleaing)**

[13]
✓ 0s

```python
# Final check
print("Shape after cleaning:", df.shape)

print("number of rows: ", df.shape[0])
print("number of columns: ", df.shape[1])
```

```
Shape after cleaning: (513, 17)
number of rows:  513
number of columns:  17
```

✏️ Explanation:

- After cleaning, there should be no missing values.
- Number of rows may decrease after removing duplicates.